5 Student X constructs a circuit to demonstrate the effect of loading a battery by adding resistors in parallel across it. An ammeter of negligible resistance measures the total current drawn from the battery.

Resistors can be added into the circuit by closing switches **A**, **B**, **C**, **D**, **E** as shown in Figure 5.1.

The battery has an e.m.f. of 9.0 V and each of the resistor has a labelled value of 10 Ω.

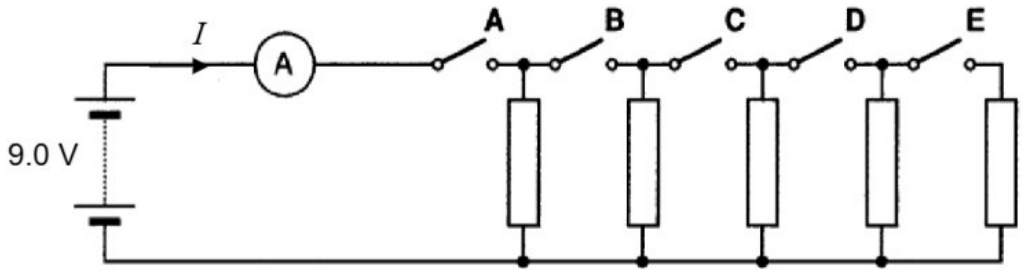


Figure 5.1

(a) Due to the way the resistors are made, there is a random variation of ± 5% in their value.

Student X assumes that the internal resistance of the battery is negligible.

Using this assumption and taking the e.m.f. of the battery to be constant, show that the possible variation in the value of the current I when only switch **A** is closed is approximately 0.05 A.

[2]

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(b) Figure 5.2 shows Student X's predicted and actual results.

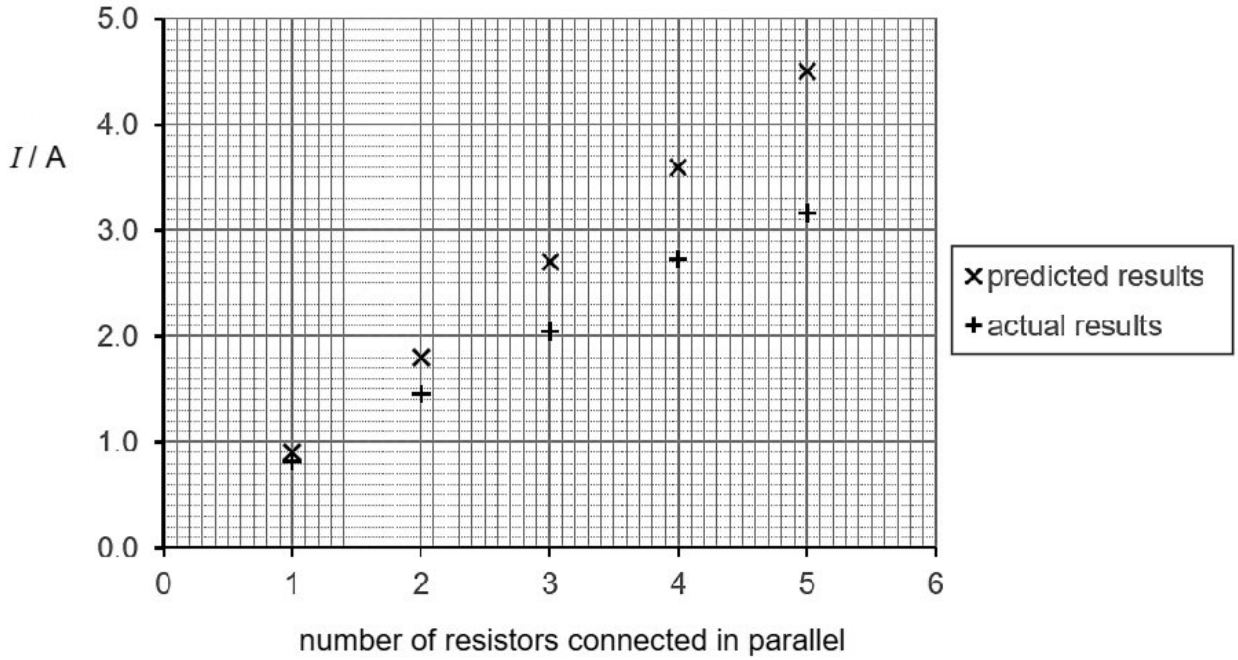


Figure 5.2

(i) Student X suggests that the difference between the two sets of results is due to the random variation (± 5%) of the value of the resistors.

Explain how the data in Figure 5.2 shows that Student X is **incorrect**.

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..... [1]

(ii) Student Y then suggests that the increasing deviation between predicted and actual results are due to the internal resistance of the 9.0 V battery.

Explain how the data in Figure 5.2 supports Student Y's statement.

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(c) Student Y then plots a graph of $1/I$ against the resistance R of the resistors in parallel to find the internal resistance of the battery. (Figure 5.3)

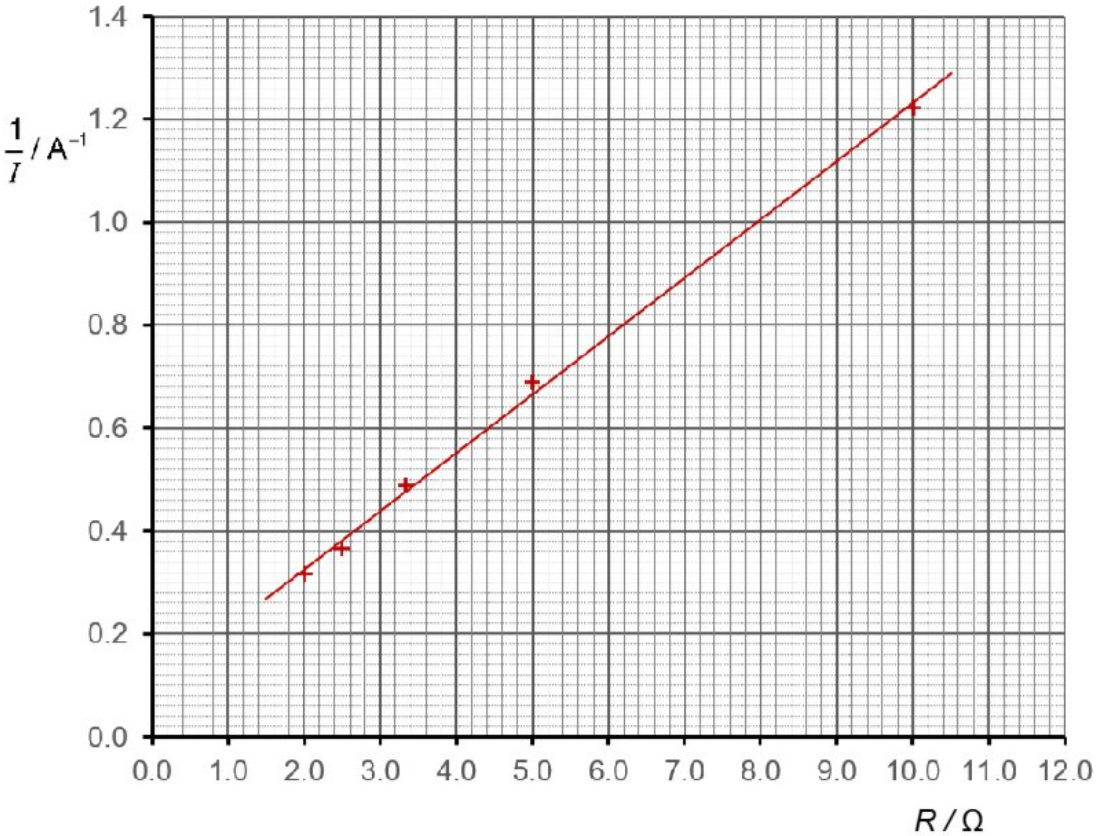


Figure 5.3

Use the graph to determine the internal resistance of the battery.

internal resistance = ... Ω [3]

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